Modulation of refractive index caused by amplitude variation and heterogeneity of anchoring forces in nanosphere-doped liquid crystal metamaterial: Monte Carlo analysis

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ABSTRACT

Recently,[1] we have discussed anchoring forces and the electric field as new control parameters for negative–positive refraction index tuning in nanosphere dispersed nematic liquid crystal (NDLC). The present study is focused on calculation of the amplitude modulation of the refractive index caused by amplitude variation of anchoring forces and spatial modulation of anchoring forces. Preliminary results indicate that, similarly to case studied earlier,[1] refractive index amplitude modulation can be significantly larger as compared with a conventional liquid crystal (LC) system. The inhomogeneous molecular order in nematic liquid crystal (NLC) cells is modelled using Monte Carlo simulations with the Lebwohl-Lasher effective Hamiltonian with the Rapini-Papoular term for anchoring forces.

Keywords: negative index of refraction, metamaterials, nematic liquid crystals, Monte Carlo

1. INTRODUCTION

Since the pioneering work of Veselago,[2] metamaterials having negative refractive indices have been proposed and demonstrated in microwave – optical region. Tunability of the refractive index of the metamaterials is an important requirement for optical applications. In the case of materials based on liquid crystals host the tunability is enabled by electrical and/or optical field controlled birefringence of the nematic liquid crystals constituents.[3–10] Nonlocal inhomogeneous nematic responses are particularly important to obtain accurate quantitative description of the index modulation. Furthermore, in cases where non-uniform fields are applied, for example, in photorefractivity-mediated optical wave mixing processes,[10] these approaches often produce new insights, and unexpected and potentially useful effects.

The aim of this paper is to calculate and analyze chosen spatial inhomogeneities of refractive index in the nano-sphere dispersed liquid crystal metamaterial[8] system, resulting from variation of anchoring forces. For this purpose, we apply Monte Carlo simulation techniques, which have been demonstrated to be well–suited for modelling such NLC responses.[11,12]

2. NANO-SPHERE DISPERSED LIQUID CRYSTAL

The presentation in this Section is based on paper.[1] In the formalism introduced by Ref.,[5,8] electromagnetic wave scattering from a single coated sphere is calculated using Mie theory[13] while the effective scattering from an ensemble of coated spheres dispersed in NLC matrix is analyzed in effective medium theory using Maxwell Garnet mixing rule.[14,15] The effective refractive index of NDLC depends on the parameters of its constitutive components: coated nonmagnetic spheres, and host NLC system (Fig. 1). Fig. 2 shows the schematic picture of a coated nano–sphere dispersed in a host nematic liquid crystal. The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

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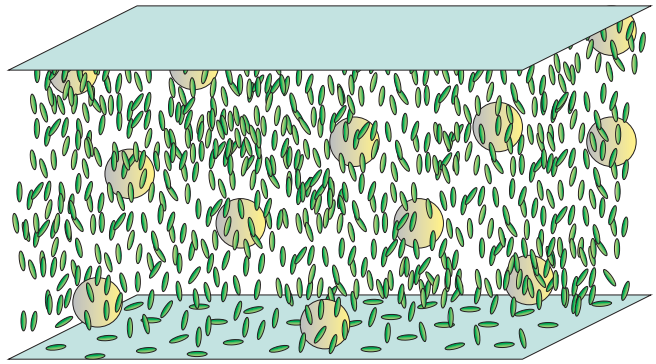

Figure 1. Nanosphere-doped liquid crystal cell.

Here $\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L stand for the transverse and longitudinal phonon frequencies, respectively. Bulk cavity resonance in the core is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths. The relative permittivity of the semiconductor shell with outer radius r_2 is described by a Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p is plasma frequency and γ_2 – the damping coefficient. The shell is responsible for negative effective permittivity, with ω_p close to frequency for which negative index will occur. Its value is chosen to provide both negative effective permittivity and permeability in some interval of wavelengths.

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle between directions of incident light wave vector and the director axis of NLC:[8–10]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are the permittivities for light polarized respectively parallel and perpendicular to the director axis. In pure NLC host (without spheres), $k_3 = \sqrt{\epsilon_3}\, k_0$.

The effective permittivity and permeability of NDLC, calculated in effective medium theory in dipolar (electric and magnetic) approximations are given by formulas:[8]

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{4}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{5}$$

More information, in particular on coefficients a_1 and b_1, can be found in.[13] These expressions are valid only for small number density $N = 3f/4\pi r_2^3$ of the nano-spheres (f is the filling factor) and in long-wavelength limit $2\pi/k_3 \gg r_2$. The effective index of refraction is

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\, \mu_{eff}}, \tag{6}$$

where we have used the root corresponding to a positive imaginary part. Based on homogeneous–response formalism, Khoo et al. have shown[8] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which the real part of effective refractive index is negative.

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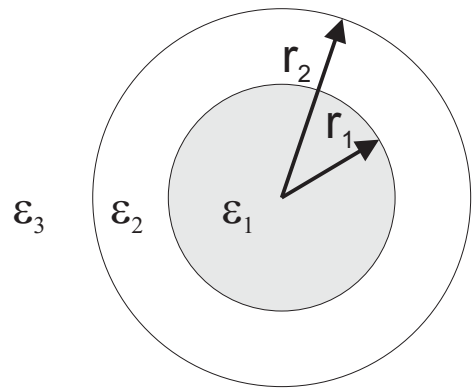

Figure 2. Coated nano-sphere.

3. MONTE CARLO SIMULATIONS AND NDLC MODEL

Monte Carlo simulations[16,17] use random sampling to explore the multi–particle configuration space to provide statistical–mechanics description of the system. Metropolis and co–workers[18] have proposed an effective sampling procedure which incorporates a temperature of the system. The method ensures that the Boltzmann average of a physical property of the system can be easily calculated. The Boltzmann distribution law states that the average of any physical property (with property value X) for a system at temperature T is given by:

$$\langle X \rangle = \frac{\sum_a X_a e^{-\frac{U_a}{k_B T}}}{\sum_a e^{-\frac{U_a}{k_B T}}}, \tag{7}$$

where the sum is over the states a available to the system. The algorithm of Metropolis is an important sampling technique. It generates a Markov chain of system states whose limiting relative frequencies are equal to their Boltzmann probabilities. In this approach sum (7) is over those states only instead over all states available to the system. In Metropolis importance sampling Monte Carlo scheme one has to: (i) start from an arbitrary initial configuration a; (ii) generate a trial configuration a'; (iii) calculate the energy difference $U_{a'} - U_a$ between trial and initial configuration a; (iv) if $U_{a'} - U_a < 0$ then the trial configuration is attached to Markov chain of configurations and algorithm returns to step (ii); (v) calculate probability $w = \exp[-(U_{a'} - U_a)/k_B T]$ and sample a random number R from interval [0,1] from uniform distribution; (vi) if $R \leq \exp(-(U_{a'} - U_a)/k_B T)$ then the trial configuration (a') is attached to Markov chain of configurations and algorithm returns to step (ii); (vii) if (vi) is not satisfied then the old configuration (a) is attached to Markov chain of configurations and the algorithm goes back to step (ii). In this paper we use above method to calculate local orientation of liquid crystal.

Practical realization of those general ideas for LC system is described in details below. We follow the argumentation presented in paper.[1] In our model the NLC molecules, represented by 1D rods, occupy sites of a $n_x \times n_y \times n_z$ rectangular lattice. Translational motions of the center of mass of the molecules are prohibited. An orientation of a long axis of a NLC molecule in an external coordinate system is described by a pair of angles ϕ, θ (Fig. 3). The angle ϕ (twist) describes an azimuth and angle θ (tilt) is a polar angle calculated from z-axis, along which the electromagnetic wave propagates with polarization along x-axis. We apply rubbing along x-axis. The molecules lie (and fluctuate) in the $x - z$ plane. Along the x and y-direction the periodic boundary conditions are applied, while the molecules in the lowest (1) and the uppermost (n_z) planes are subjected to properly chosen anchoring forces due to their interaction with a substrate. Interactions are described by the Lebwohl-Lasher effective Hamiltonian[19] with Rapini-Papoular term[20] in units $k_B T$ (k_B – Boltzmann constant, T – absolute temperature):

$$H = -\xi \sum_{<\vec{r},\vec{r}\,'>} P_2(\cos\beta(\vec{r},\,\vec{r}\,')) - \sum_{\vec{r}} E^2(\vec{r}))\, P_2(\cos\beta(\vec{r})) + \sum_{\vec{r}_W} \alpha(\vec{r}_W)\, \sin^2\gamma(\vec{r}_W), \tag{8}$$

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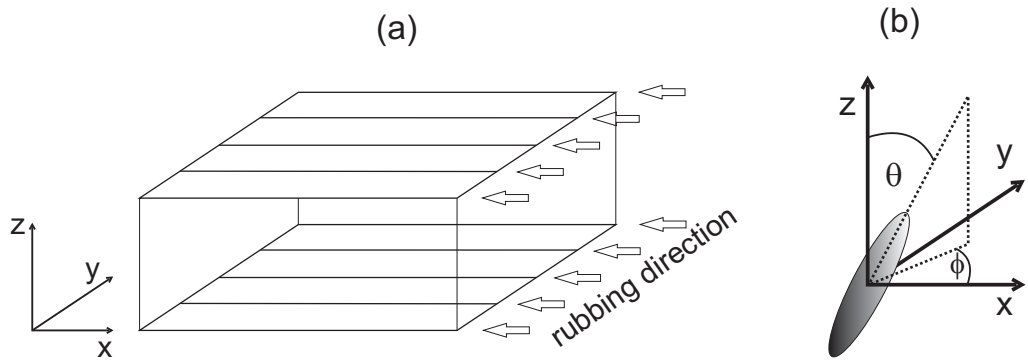

Figure 3. Geometry of the system (a) and orientation of LC molecule (b).

Here P_2 is the second-order Legendre polynomial. Parameter ξ is the coupling constant for NLC-NLC orientational interaction (in simulations we put $\xi = 25$), $E(\vec{r})$ is proportional to external electric field, $\alpha(\vec{r}_W)$ denotes the amplitude of anchoring forces acting on molecules located at the walls. Parameters ξ, α, E^2 are given in units of k_BT. $\beta(\vec{r}, \vec{r}\,')$ denotes the relative angle between two molecules located at points $\vec{r}, \vec{r}\,'$. $\beta(\vec{r})$ is the angle a molecule located at point $\vec{r}$ makes with the electric field direction. $\gamma(\vec{r}_W)$ denotes the angle between long axis of molecule at the wall and a fixed rubbing direction (Fig. 3). Each molecule in the bulk is surrounded by the six nearest neighbors and interactions only with those neighbors are taken into account.

The equilibrium configurations of NLC molecules on a lattice were sampled using Metropolis Monte Carlo simulations.[16,17] The system was equilibrated during 3×10^4 MC steps (MCS) and then simulated during 2×10^5 MCS. From 2×10^5 equilibrium configurations generated during simulation 2×10^2 configurations (separated by 10^3 MCS) were used to calculate the averaged values of chosen physical properties. In the simulations we have used the parameters from Ref. 8: $r_1 = 0.13\ \mu\mathrm{m}, \mathrm{r}_2 = 0.143\ \mu\mathrm{m}, \mathrm{f} = 0.12, \epsilon(\infty) = 17, \omega_\mathrm{T}/2\pi = 240$ THz, $\omega_L/2\pi = 570$ THz, $\gamma_1/2\pi = 2.5$ THz, $\omega_\mathrm{p}/2\pi = 134$ THz and $\gamma_2 = \omega_p/60$.

4. HOMOGENEOUS ANCHORING STRENGTH AND SPATIALLY MODULATED ELECTRIC FIELD

In this case spatially modulated electric field takes the form:

$$E = E_{DC} + A\sin(qx). \tag{9}$$

It consists of constant part E_{DC} and of modulated part with amplitude $A = E_{DC}/10$. In Fig. 4 inhomogeneous spatial distribution of real part of refractive index $n'_{eff}(x, z)$ in NDLC resulting from modulation of electric field for $\lambda = 2830$ nm is presented. Simulations were performed for fixed values of electric field ($E_{DC} = 0.8, A = 0.08$) and for strong ($\alpha = 50$), middle ($\alpha = 30$) and weak ($\alpha = 10$) homogeneous anchoring energy. In the cases of strong and medium anchoring regions of metamaterial and normal material are observed. When the anchoring is weak then real part of effective refractive index is negative almost in a whole sample.

In the next step averaged index $\langle n'_{eff}\rangle(x)$, which is related to the phase shift of light transmitted through cell of thickness L along z-direction, was calculated according to formula:

$$\langle n'_{eff}\rangle(x) = \frac{1}{L}\int_0^L n'_{eff}(x, z)dz. \tag{10}$$

Amplitude of this averaged value is directly related to diffraction efficiency η which is proportional to index amplitude squared:

$$\eta \propto (\Delta\langle n'_{eff}\rangle)^2. \tag{11}$$

In Fig. 5 (left) we show the profiles $\langle n'_{eff}\rangle(x)$. For used parameters we have modulation of refractive index with $\langle n'_{eff}\rangle > 0$ for strong anchoring and with $\langle n'_{eff}\rangle < 0$ for weak anchoring. For those both cases amplitude of

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modulation is approximately equal to 0.5 (Fig. 5 (right)) and the modulation of $\langle n'_{eff} \rangle$ is opposite to modulation of electric field. The results imply the existence of a maximum of index modulation for intermediate anchoring conditions. Moreover, the index amplitude modulation is a few times larger than in a conventional LC system.

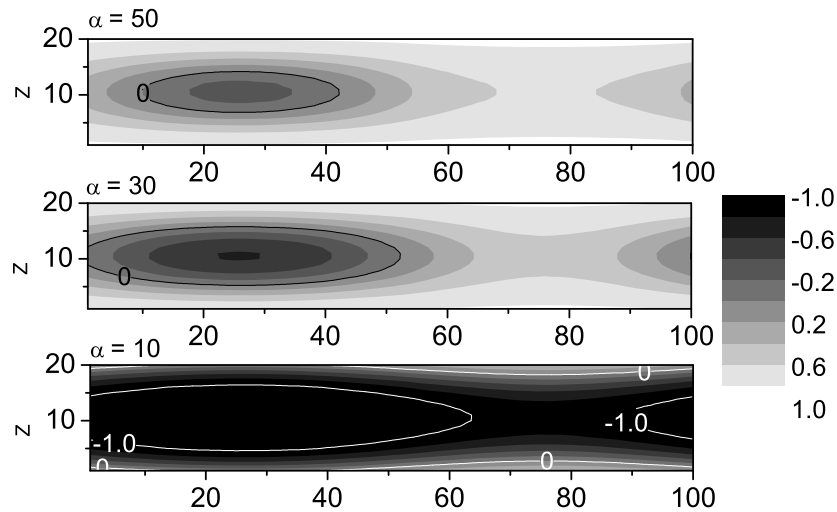

Figure 4. 2D maps of distribution of $n'_{eff}(x, y, z)$ averaged over y axis for $\lambda = 2830$ nm($E_{DC} = 0.8, A = 0.08$) for strong ($\alpha = 50$), middle ($\alpha = 30$) and weak ($\alpha = 10$) homogeneous anchoring energy.

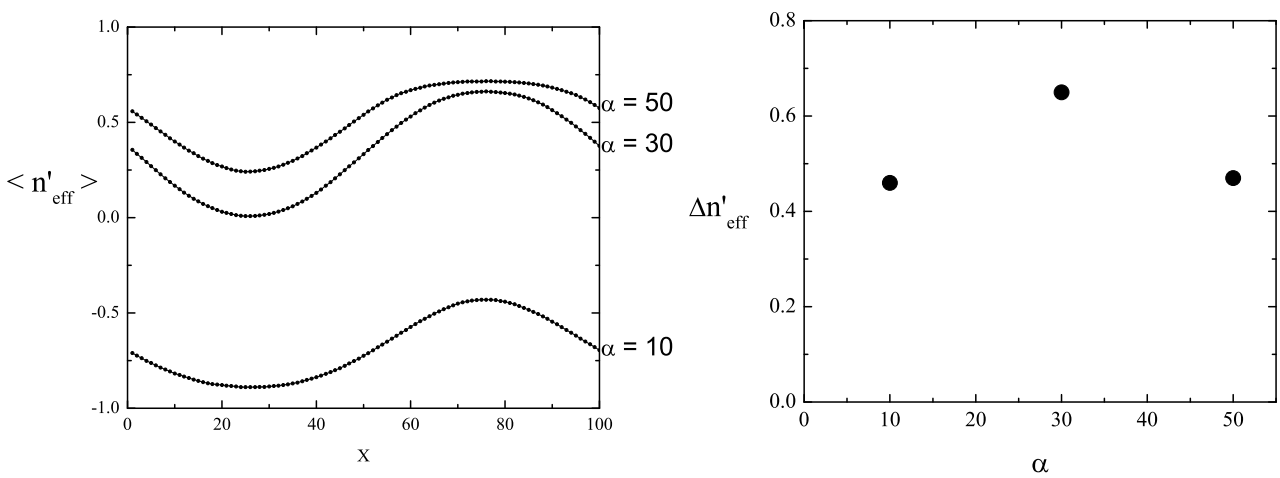

Figure 5. Profiles $\langle n'_{eff} \rangle(x)$ calculated for data from Fig. 4 (left) and amplitudes of modulation of $\langle n'_{eff} \rangle$ (right) for different values of anchoring energy.

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5. SPATIALLY MODULATED ANCHORING STRENGTH AND HOMOGENEOUS ELECTRIC FIELD

Inhomogeneity of anchoring can be as important as inhomogeneity of external electric field. Below, we investigate the simplest case when inhomogeneity is described by step function. Regions of surface with strong anchoring ($\alpha = 50$) are placed close to regions without anchoring ($\alpha = 0$) as shown schematically in Fig. 6). Anchoring

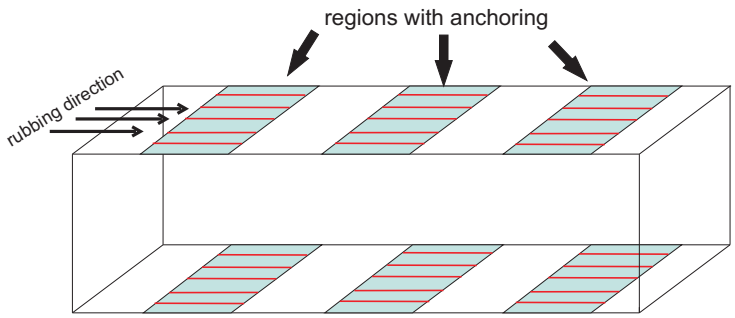

Figure 6. LC cell with modulation of anchoring.

energy reads:

$$\alpha(x) = \begin{cases} 50 & x \leq 50 \\ 0 & x > 50 \end{cases} .$$

Profile of anchoring energy $\alpha(x)$ for one period of modulation is presented in Fig. 7. In this section homogeneous electric field is used ($A = 0$).

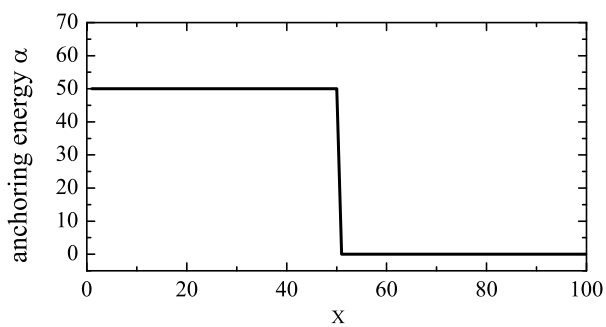

Figure 7. Profile of anchoring energy along x axis for one modulation period.

In Fig. 8 2D maps of local refractive index for $\lambda = 2820$ nm and electric field $E = 0.5$ and $E = 0.75$ are presented. Those preliminary results of simulations show that large difference between refractive index in neighboring places can be easily created in such systems. There are no horizontal (along x axis) layered structures which are typical of systems with modulated field.

6. CONCLUSIONS

We have presented a quantitative analysis of the effect of an applied spatially modulated field and spatially modulated anchoring energy on a metamaterial comprising nano-spheres dispersed in an aligned nematic liquid crystal. We have shown first results which indicate that amplitude of refractive index can be much larger than that for a conventional LC system.

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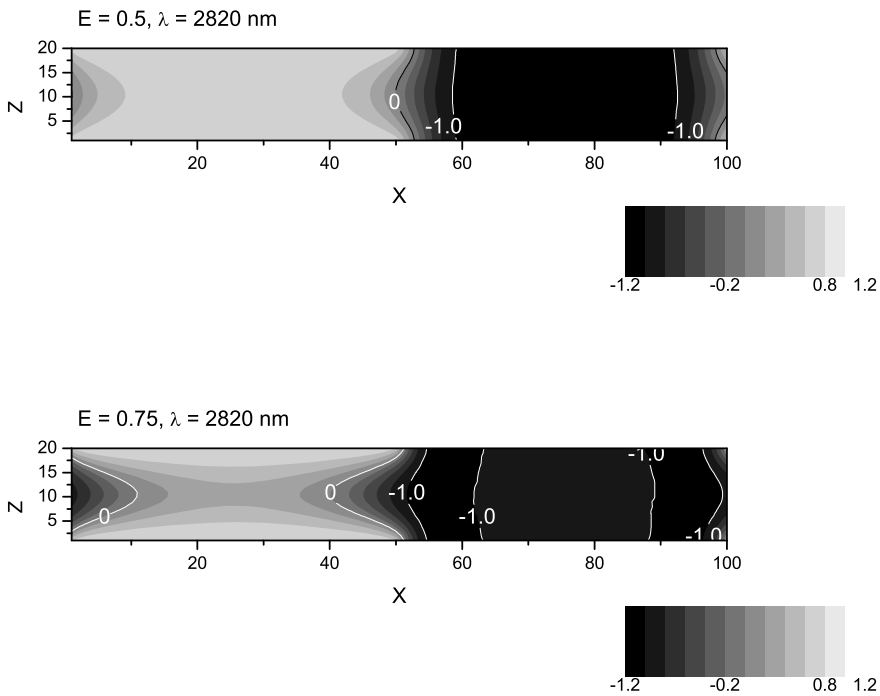

Figure 8. 2D maps of distribution of $n'_{eff}(x, y, z)$ averaged over y axis for $\lambda = 2820$ nm, $E = 0.5$ (top) and $E = 0.75$ (bottom).

REFERENCES

1. G. Pawlik, M. Jarema, W. Walasik, A. C. Mitus, and I. C. Khoo, J. Opt. Soc. Am. B **27**, 567 (2010).
2. V. G. Veselago, Sov. Phys. Usp. **10**, 509 (1968).
3. J. A. Bossard, X. Liang, L. Li, D. H. Werner, B. Weiner, P. F. Cristman, A. Diaz, I. C. Khoo, IEEE Transactions on Antennas and Propagation **56**, 1308 (2008).
4. X. Wang, D. H. Kwon, D. H. Werner, I. C. Khoo, A. Kildishev and V. M. Shalaev, Appl. Phys. Lett. **91**, 143122 (2007).
5. M. S. Wheeler, J. S. Aitchison and M. Mojahedi, Phys. Rev. B **73**, 045105 (2006).
6. Q. Zhao, L. Kang, B. Du, B. Li, J. Zhou, H. Tang, X. Liang, B. Z. Zhang, Appl. Phys. Lett. **90**, 011112 (2007).
7. I. C. Khoo, A. Diaz, S. Kubo, J. Liou, Mike Stinger, T. Mallouk and J. H. Park, Molecular Crystals Liquid Crystals **485**, 934 (2008).
8. I.C. Khoo, D.H. Werner, X. Liang, and A. Diaz, Opt. Lett. **31**, 2592 (2006).
9. I. C. Khoo, *Liquid Crystals*, 2nd ed., Wiley, NJ 2007.
10. I. C. Khoo, Physics Report **471**, 221 (2009).
11. G. Pawlik, A.C. Mitus, and A. Miniewicz, Opt. Commun. **182**, 249 (2000).
12. K. Komorowska, G. Pawlik, A.C. Mitus, and A. Miniewicz, J. Appl. Phys. **90**, 1836 (2001).
13. C.F. Bohren and D.R. Huffman, Absorption and Scattering of Light by Small Particles, (Wiley, 1998).
14. J.C. Maxwell Garnett, Philos. Trans. R. Soc. Ser. A **203**, 385 (1904).
15. C. A. Gurin, P. Mallet, and A. Sentenac, J. Opt. Soc. Am. A **23**, 349 (2006).
16. P. Allen and D. Tildesley, *Computer Simulation of Liquids*, Clarendon Press, Oxford, 1987.
17. D. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge University Press, 2000.

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18. N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller, J. Chem Phys. **21**, 1087 (1953).
19. P.A. Lebwohl and G. Lasher, Phys. Rev. A **6**, 426 (1972).
20. A. Rapini and M. Papoular, J. Phys. (France) **30**, 54 (1969).

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